

- Describe **and** explain how the use of a marker gene coding for a fluorescent product can show that the introduced gene of interest is being expressed in plants.

(b) Give **two** advantages of genetically engineering herbicide resistance in crop plants such as soybean.

..... [2]

- (c) DNA microarray analysis is a technique used in genetic technology that involves fluorescence. A DNA microarray has single-stranded probes attached to its surface. These probes hybridise to the fluorescently tagged single-stranded DNA that is added.

- (i) Explain why DNA hybridisation occurs between the probe DNA and the added DNA.

.....
..... [1]

- (ii) A DNA microarray analysis can be used to identify the level of expression of some genes.

Describe **and** explain how the level of expression of some genes can be identified using the DNA microarray analysis technique.

.....
.....
.....
.....
.....
.....
.....
..... [3]

[Total: 10]